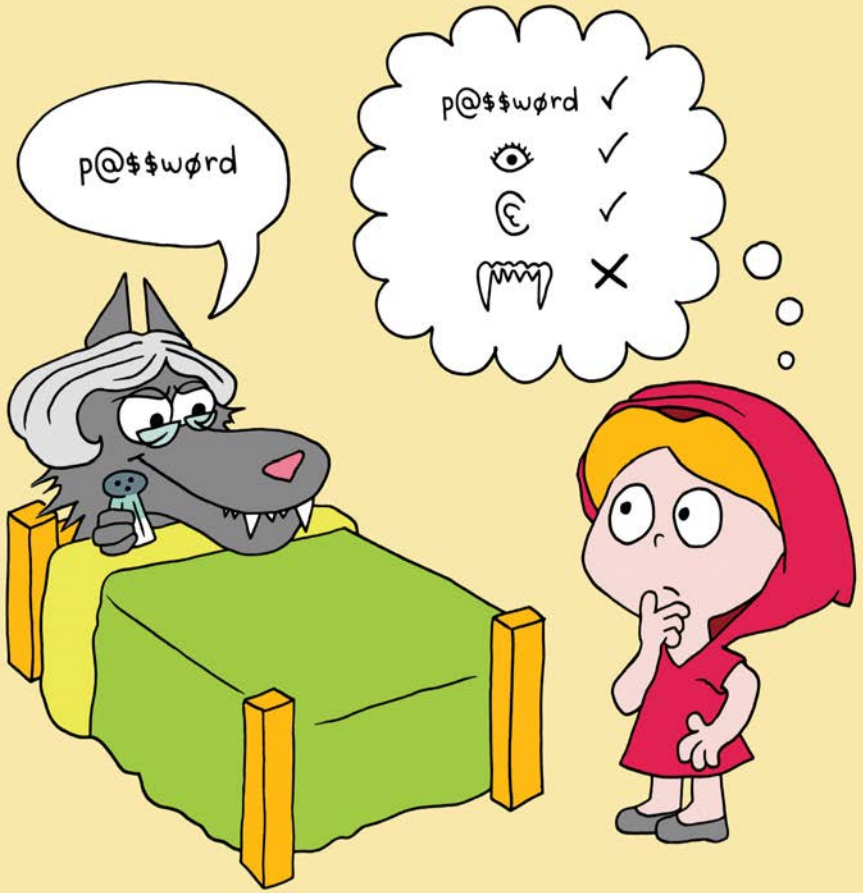
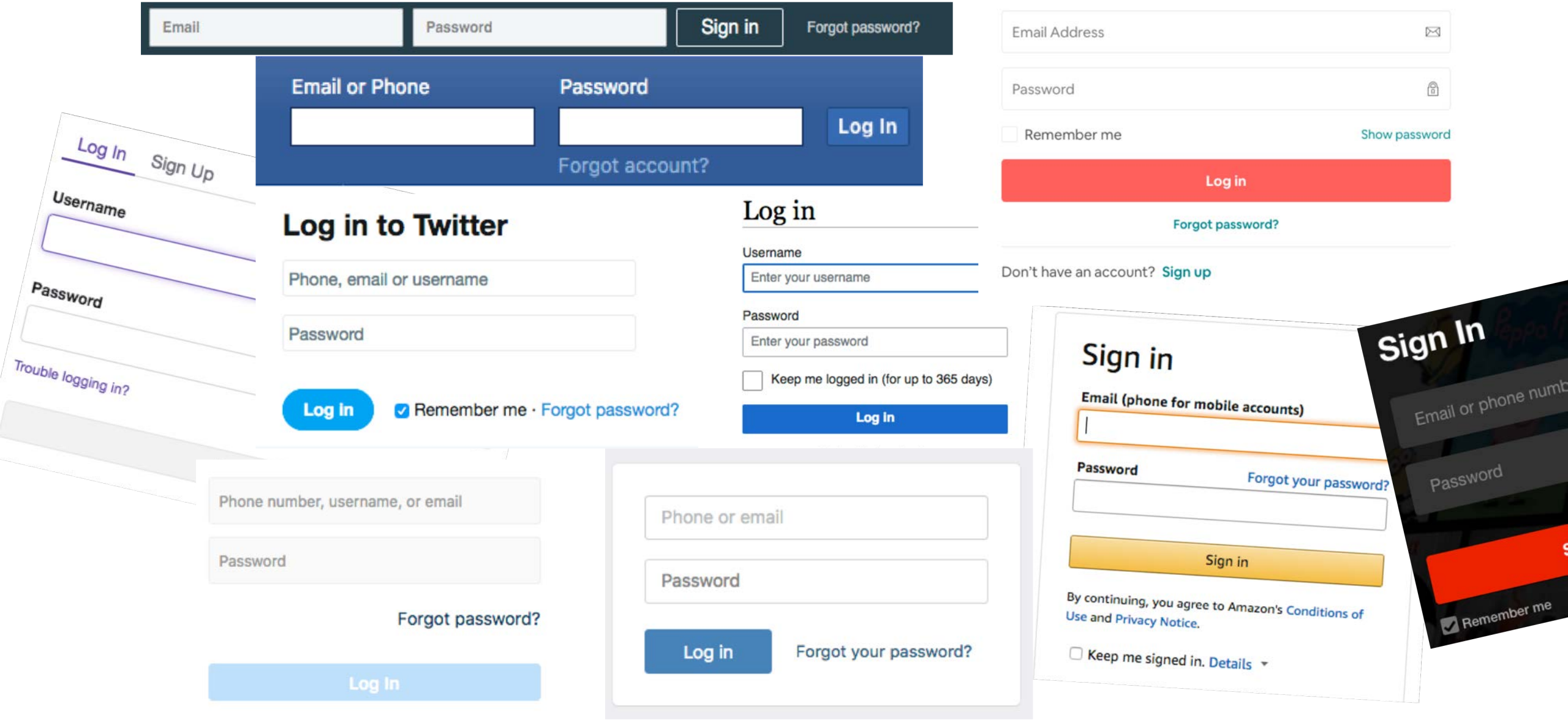


Image: Iwona Usakiewicz / Andrij Borys Associates
Bonneau et al.: Passwords and the evolution of imperfect authentication. Comm. ACM 58(7) (Jun 2015)

Is This Really You? An Empirical Study on Risk-Based Authentication Applied in the Wild

Stephan Wiefling, Luigi Lo Iacono – TH Köln – University of Applied Sciences
Markus Dürmuth – Ruhr University Bochum


Email
Password
Sign in
Forgot password?
Email or Phone
Password
Log In
Forgot account?
Log In
Sign Up
Username
Password
Trouble logging in?
Log in to Twitter
Phone, email or username
Password
Log In
Remember me · Forgot password?
Log in
Username
Enter your username
Password
Enter your password
Keep me logged in (for up to 365 days)
Log In
Email Address
Password
Remember me
Show password
Log in
Forgot password?
Don't have an account? Sign up
Phone number, username, or email
Password
Forgot password?
Log In
Phone or email
Password
Log in
Forgot your password?
Sign in
Email (phone for mobile accounts)
Password
Forgot your password?
Sign in
By continuing, you agree to Amazon's Conditions of Use and Privacy Notice.
Keep me signed in. Details
Sign In
Email or phone numb
Password
Remember me

Motivation

- Weaknesses in password-based authentication increase
  - Large-scale password database leaks
    - Credential Stuffing
  - Intelligent password guessing*
  - Phishing

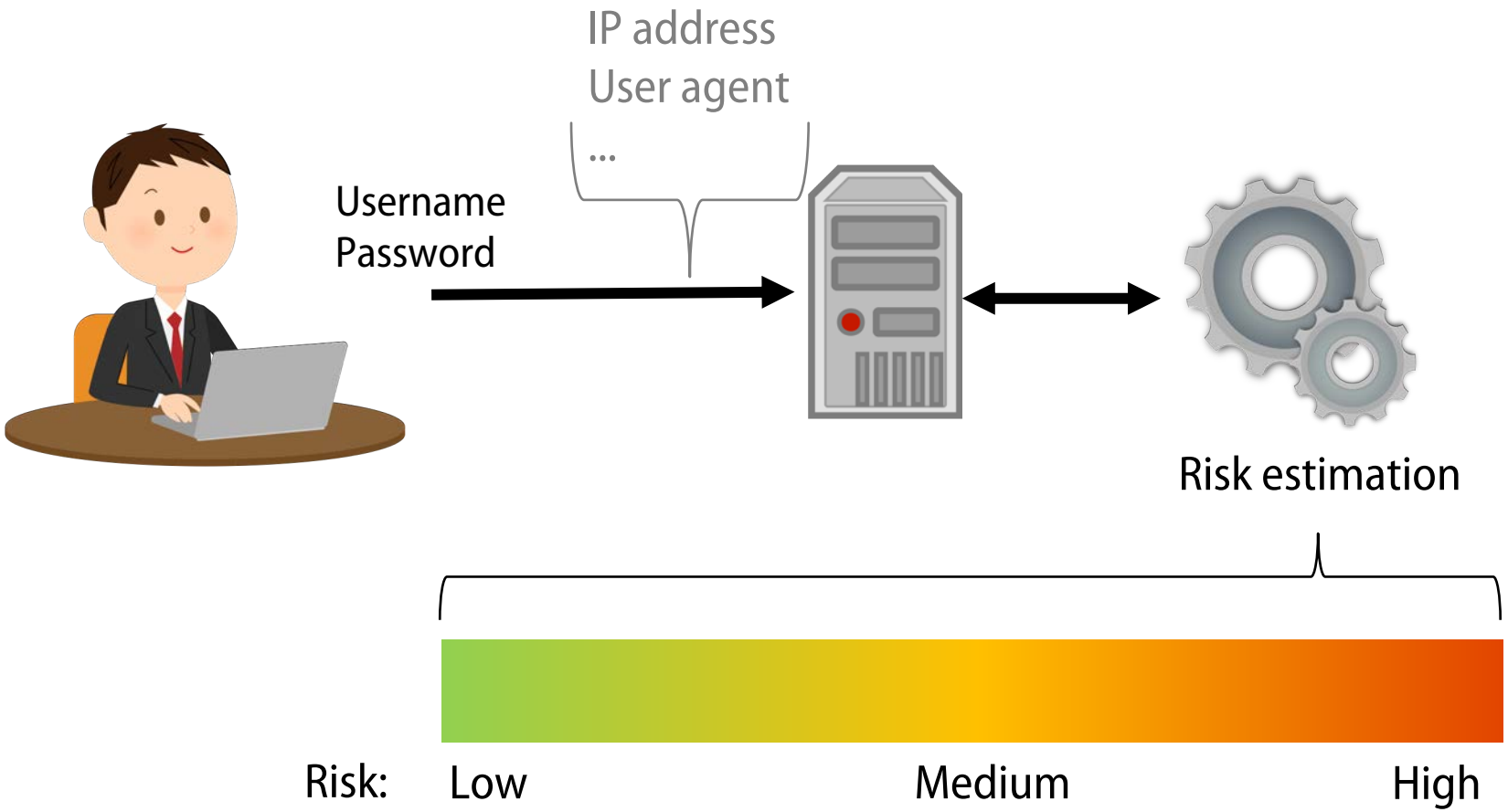
*Wang et al.: Targeted online password guessing: An underestimated threat. In CCS '16. ACM (2016)

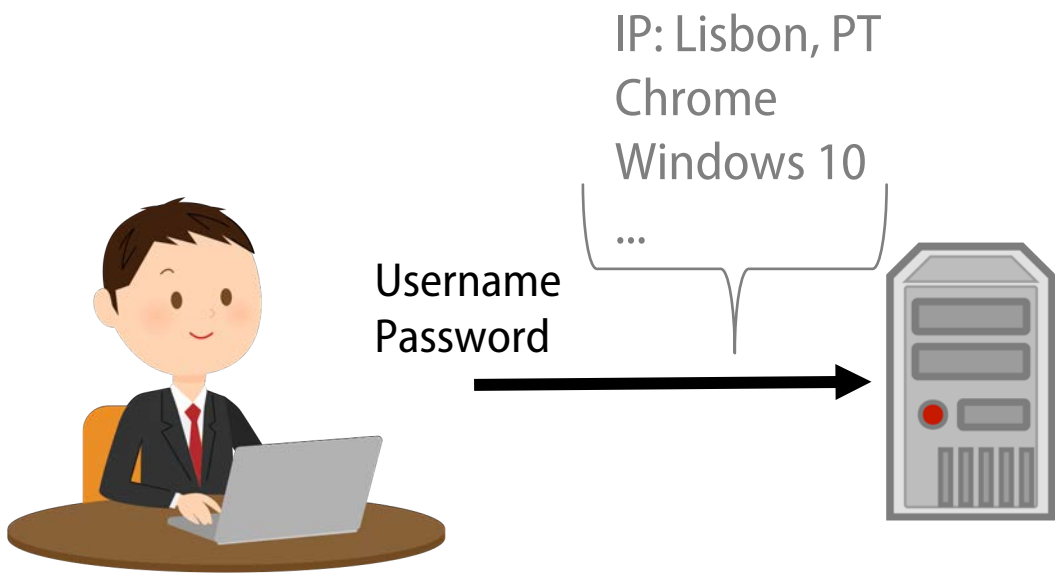
Motivation

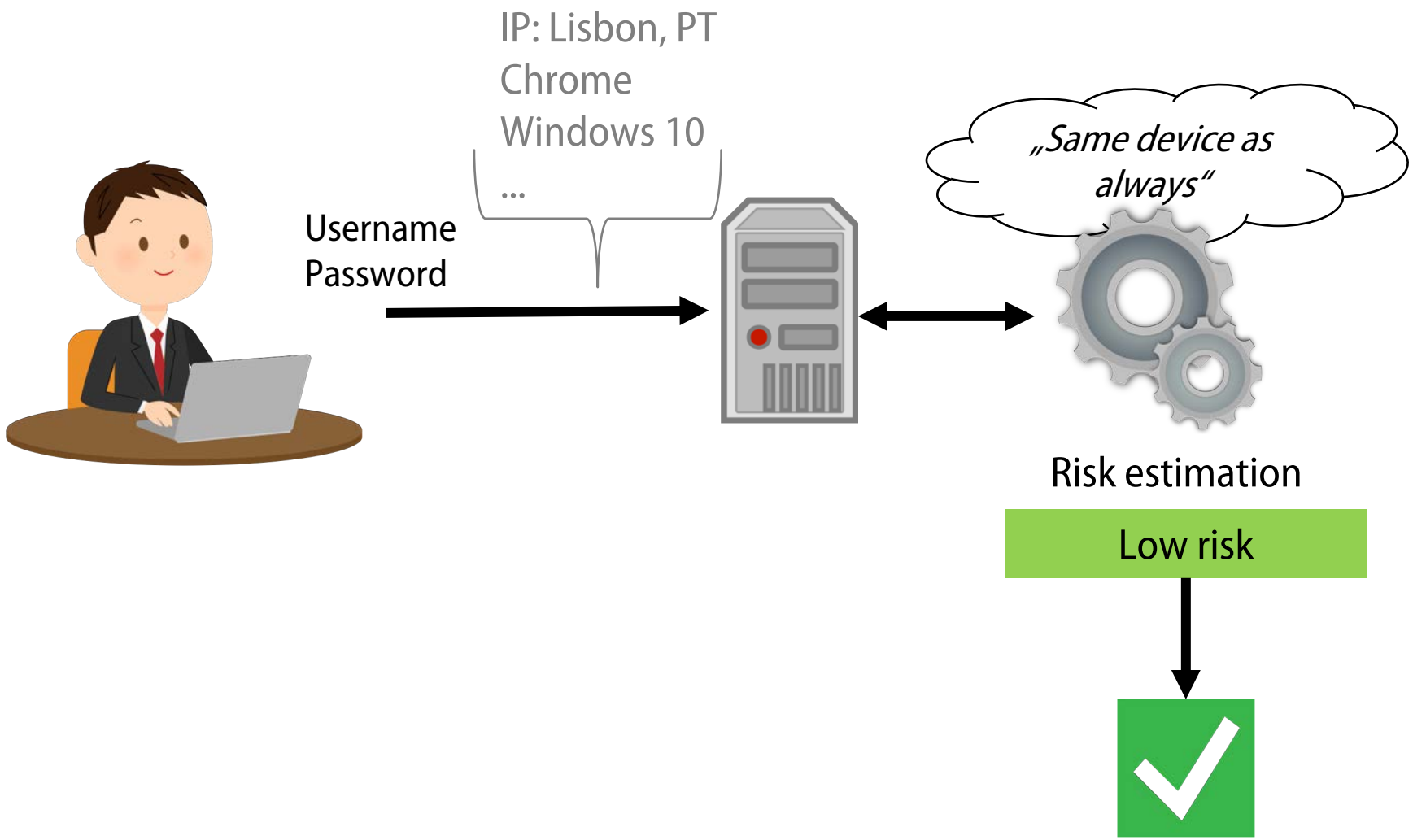
- 2FA is unpopular
  - <10% of all Google accounts used 2FA in January 2018*

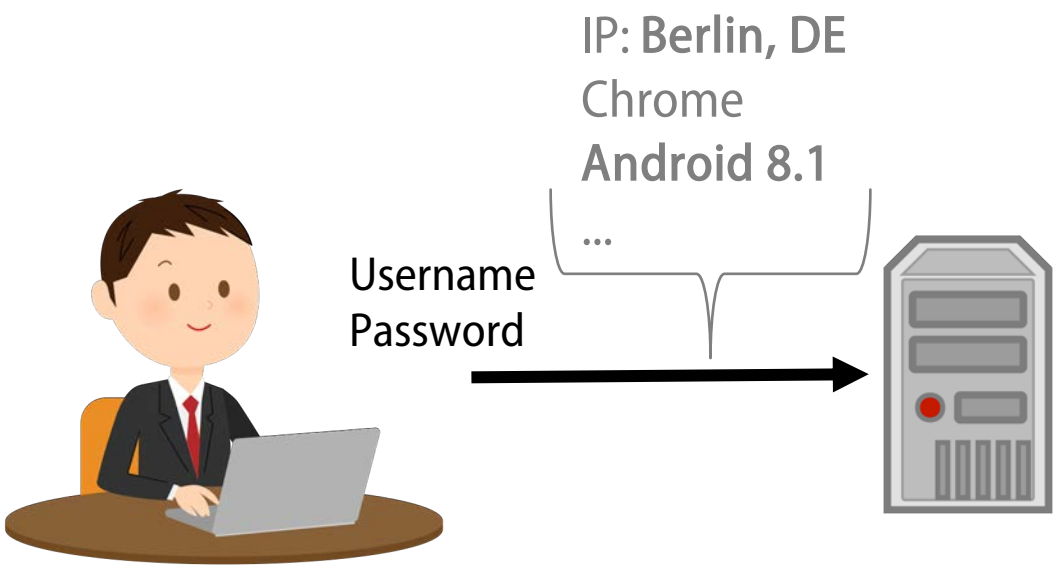
→ Using Risk-based Authentication
to increase account security
with minimal impact on user interaction

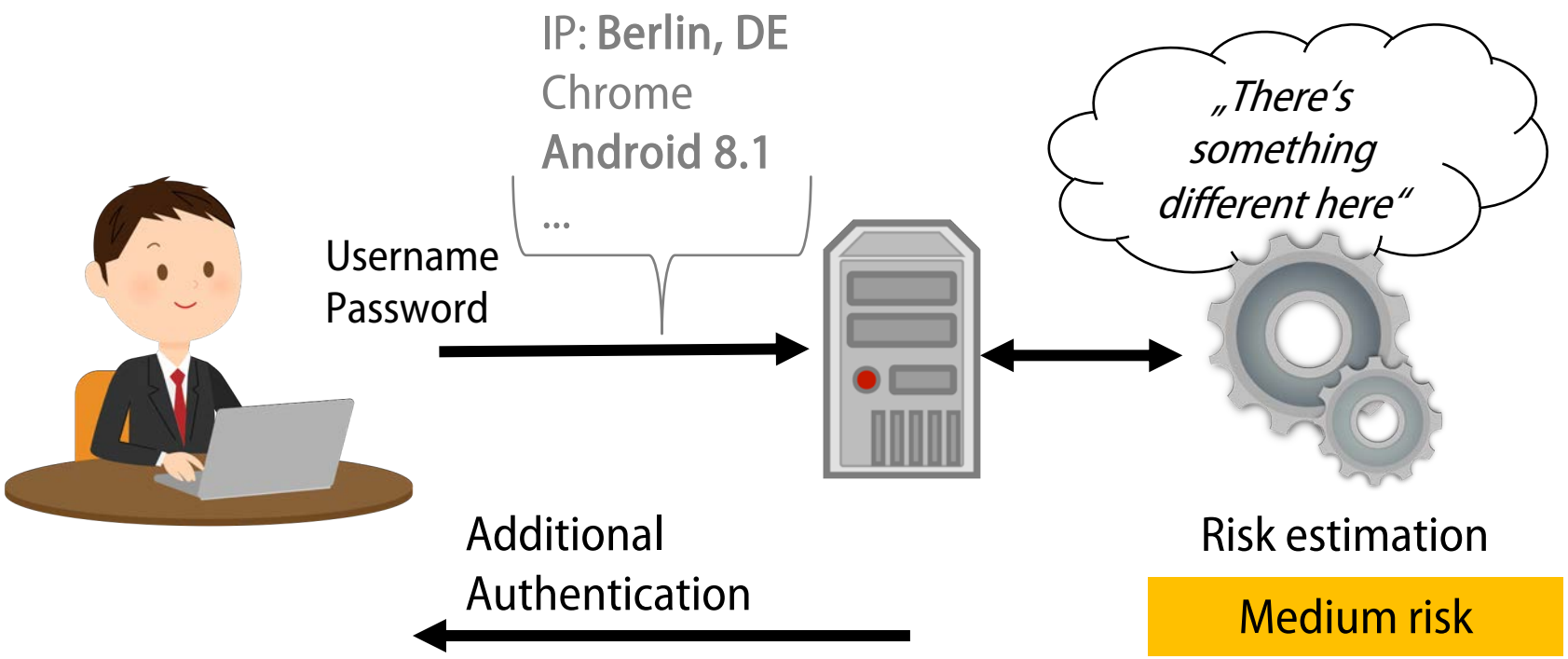
*Milka, G.: Anatomy of Account Takeover. In: Enigma 2018. USENIX (Jan 2018)

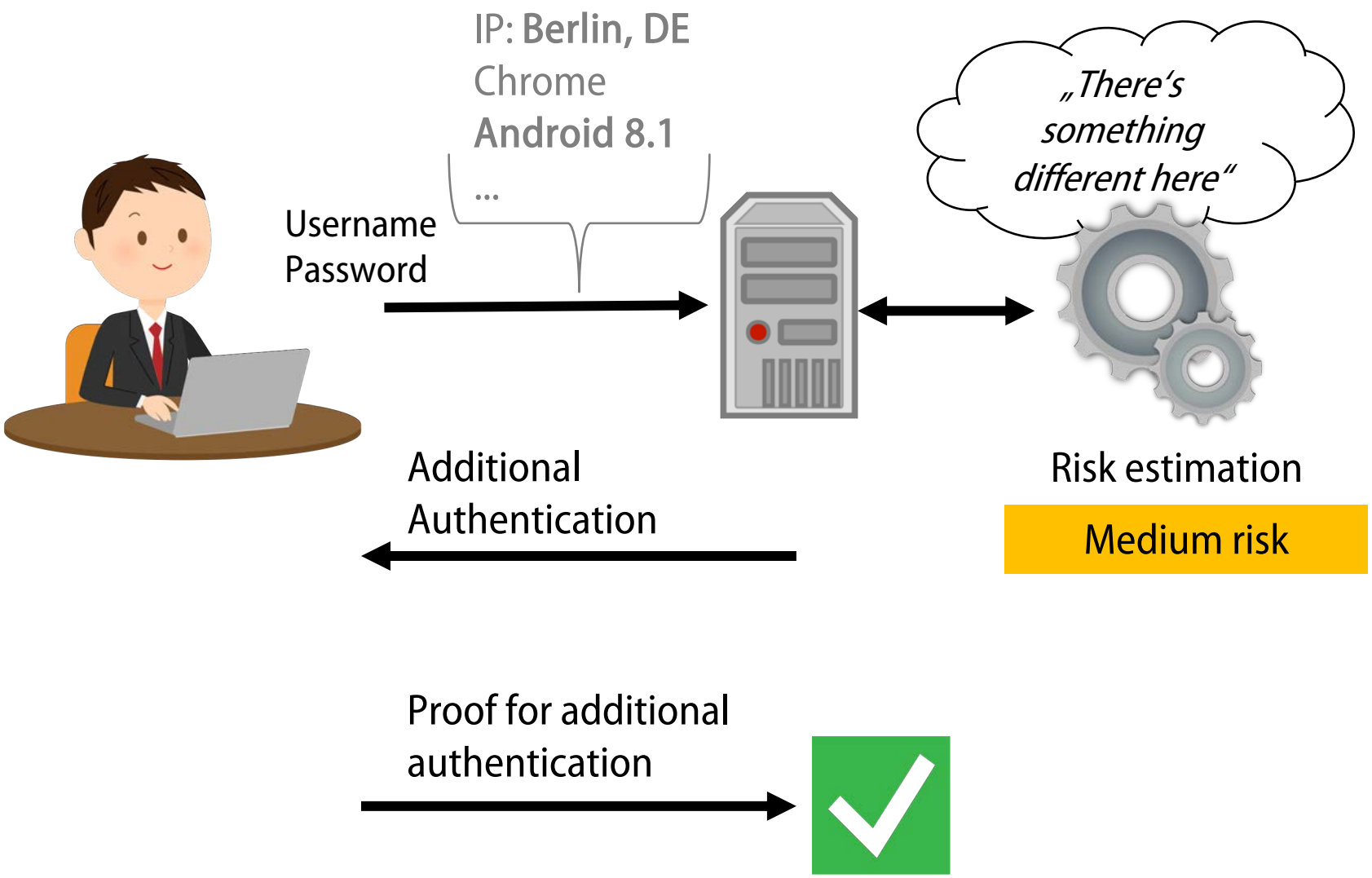

IP address
User agent
...
Username
Password
Risk estimation
Risk:
Low
Medium
High

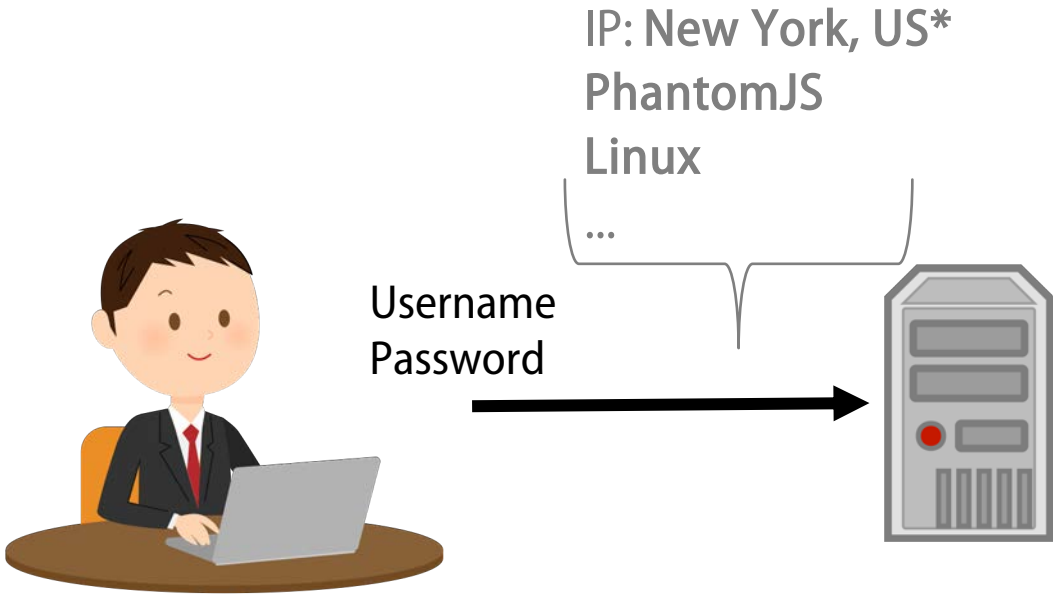

IP: Lisbon, PT
Chrome
Windows 10
...
Username
Password


IP: Lisbon, PT
Chrome
Windows 10
...
Username
Password
„Same device as always“
Risk estimation
Low risk

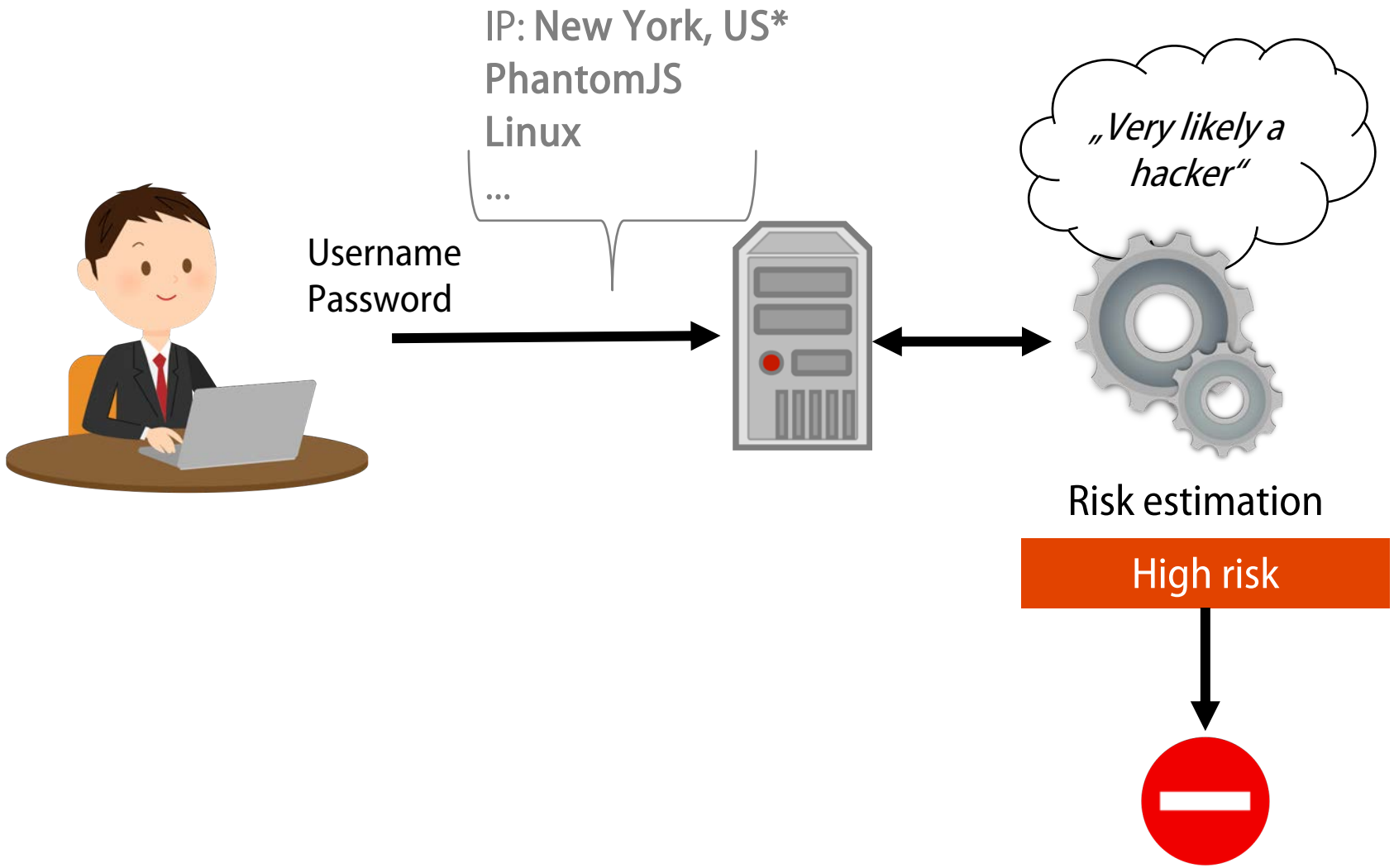

IP: Berlin, DE
Chrome
Android 8.1
...
Username
Password


IP: Berlin, DE
Chrome
Android 8.1
...
Username
Password
„There's something different here"
Additional Authentication
Risk estimation
Medium risk


IP: Berlin, DE
Chrome
Android 8.1
...
Username
Password
„There's something different here"
Additional Authentication
Risk estimation
Medium risk
Proof for additional authentication



*Known spam IP address



*Known spam IP address

Risk-based Authentication

- Recommended by NIST digital identity guidelines*
- Used by large online services
- However: Procedures not disclosed

*Grassi et al.: Digital identity guidelines. Tech. Rep. NIST SP 800-63b (2017)

Risk-based Authentication

- Recommended by NIST digital identity guidelines*
- Used by large online services
- However: Procedures not disclosed
  - Prevents widespread adoption

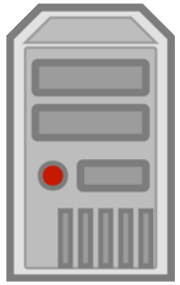
*Grassi et al.: Digital identity guidelines. Tech. Rep. NIST SP 800-63b (2017)

Risk-based Authentication

- Recommended by NIST digital identity guidelines*
- Used by large online services
- However: Procedures not disclosed

→ Black-box testing eight popular online services

*Grassi et al.: Digital identity guidelines. Tech. Rep. NIST SP 800-63b (2017)



?

Login	IP address	User Agent	...

Login →

Login	IP address	User Agent	...
1	TH Köln	Chrome	...

Login →

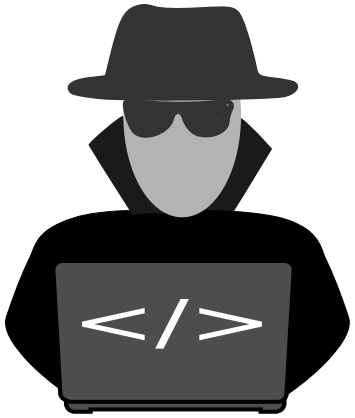
Login	IP address	User Agent	...
1	TH Köln	Chrome	...
2	TH Köln	Chrome	...

Login →

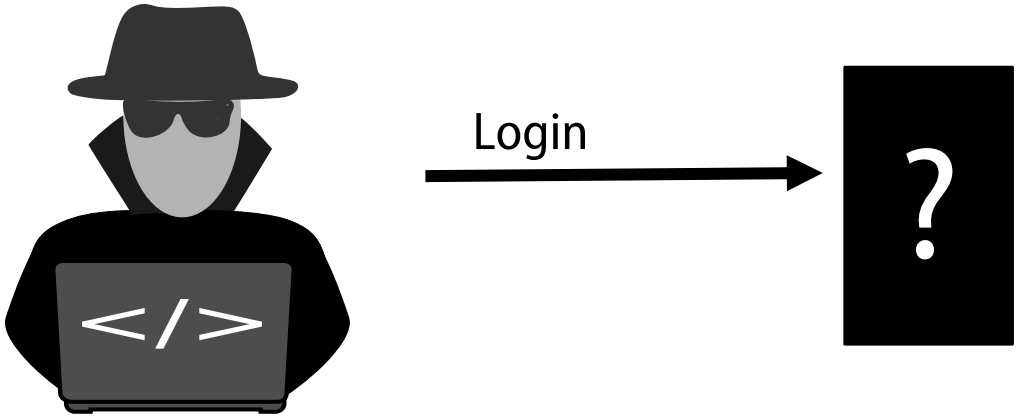
Login	IP address	User Agent	...
1	TH Köln	Chrome	...
2	TH Köln	Chrome	...
3	TH Köln	Chrome	...

Login →

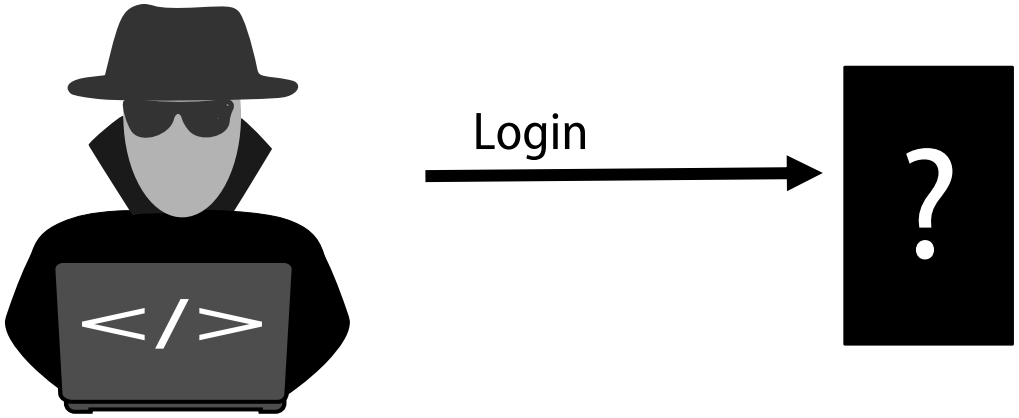
Login	IP address	User Agent	...
1	TH Köln	Chrome	...
2	TH Köln	Chrome	...
3	TH Köln	Chrome	...
...	...	...	....
20	TH Köln	Chrome	...



Login	IP address	User Agent	...
1	TH Köln	Chrome	...
2	TH Köln	Chrome	...
3	TH Köln	Chrome	...
...	...	...	....
20	TH Köln	Chrome	...



Login	IP address	User Agent	...
1	TH Köln	Chrome	...
2	TH Köln	Chrome	...
3	TH Köln	Chrome	...
...	...	...	....
20	TH Köln	Chrome	...
21	**Other Country**	Chrome	...



Login	IP address	User Agent	...
1	TH Köln	Chrome	...
2	TH Köln	Chrome	...
3	TH Köln	Chrome	...
...	...	...	....
20	TH Köln	Chrome	...
21	**Other Country**	Chrome	...

It‘s not that easy...

Login history influences risk score

It‘s not that easy...

Login history influences risk score

Solution: Create many user accounts

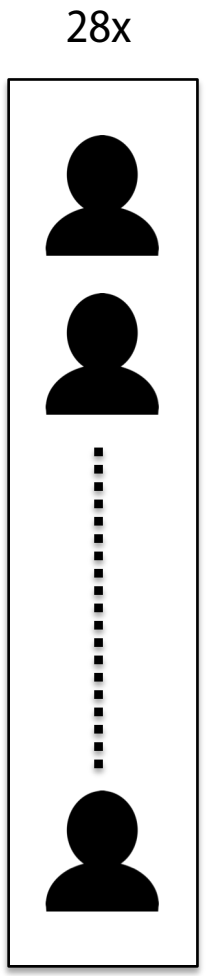
It‘s not that easy...

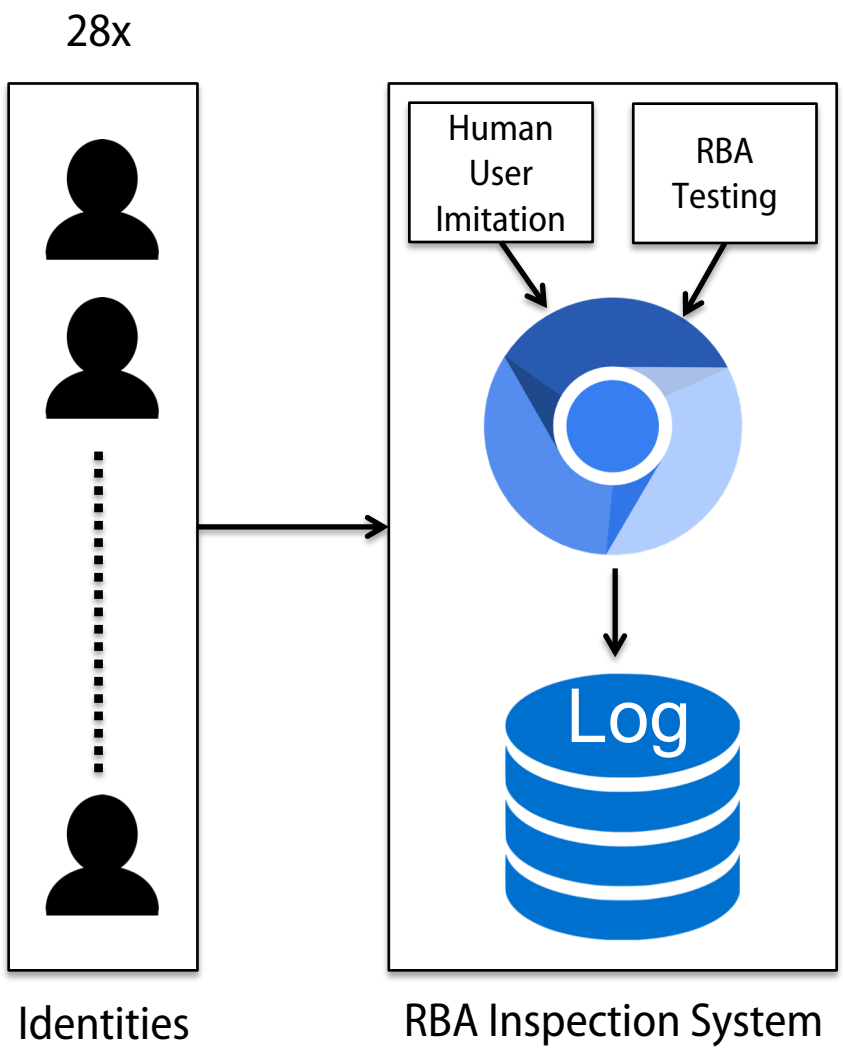
Automated testing influences result

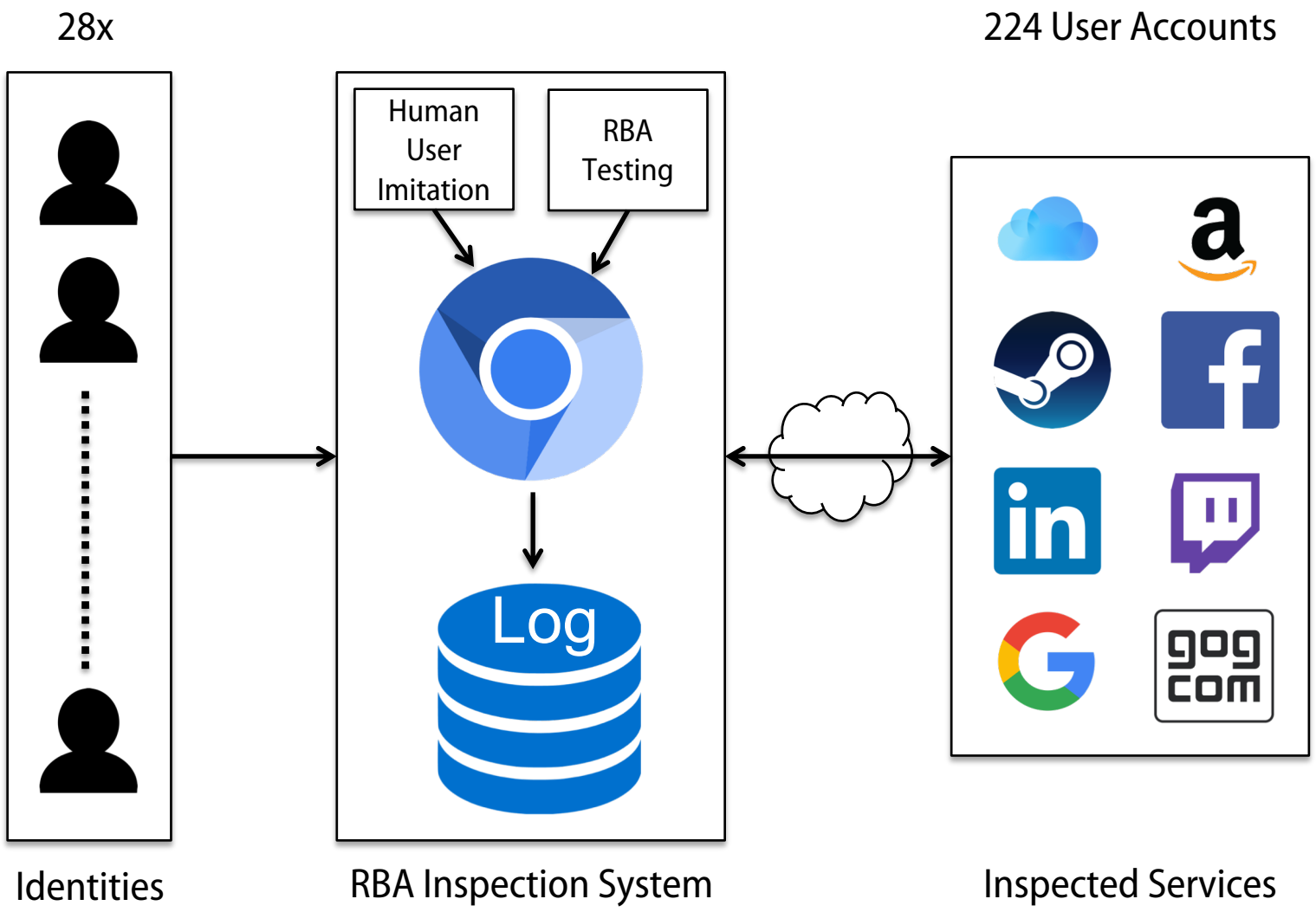
It‘s not that easy...

Automated testing influences result

Solution: Create human-like browsing behavior


28x
Identities


28x
Human User Imitation
RBA Testing
Log
Identities
RBA Inspection System


28x
224 User Accounts
Human User Imitation
RBA Testing
Log
Identities
RBA Inspection System
Inspected Services

It‘s still not that easy...

List of potential features is huge

It‘s still not that easy...

List of potential features is huge

Solution: Test most relevant* features

*Citations in literature, Highest distinguishing info in Alaca and van Oorschot

Alaca, F., van Oorschot, P.C.: Device Fingerprinting for augmenting web authentication. In: Proc. ACSAC '16. pp. 289-301. ACM (2016)

It's still not that easy...

Feature	RBA references count (except *)	Distinguishing info*
IP address	■■■■■■	●●●●○
User agent string	■■■	●●●●○
Language	■■■	●●●●○
Display resolution	■■	●●●●○
Login time	■■■■■	●●●○○
Evercookies	■	●●●●●
Canvas fingerprinting	■■■	●●●○○
Mouse and keystroke dynamics	■	-
Failed login attempts	■■	-
WebRTC	-	●●●○○
Counting hosts behind NAT	-	●●○○○
Ad blocker detection	-	●○○○○

*Alaca, F., van Oorschot, P.C.: Device Fingerprinting for augmenting web authentication. In: Proc. ACSAC '16, pp. 289-301. ACM (2016)

It‘s still not that easy...

List of potential features is huge

Solution: Test most relevant features

- IP address
- User agent string
- Language
- Display resolution
- Login time

It‘s still not that easy...

IP address feature has wide range of values

It‘s still not that easy...

IP address feature has wide range of values

Solution: Conduct a two part study

It‘s still not that easy...

IP address feature has wide range of values

Solution: Conduct a two part study

1. Find IP feature thresholds
2. Test all features with the IP threshold

It‘s still not that easy...

Study one

Find IP feature thresholds

Results

IP variation	Facebook	Google	Amazon	LinkedIn	GOG.com	Steam	Twitch	iCloud
#0 (TH Köln, fixed)	-	-	-	-	-	-	-	-
#1 (TH Köln, fresh)	-	-	-	-	A	-	-	-
#2 (same city, different ISP)	-	S	-	-	A	-	-	-
#3 (Frankfurt, DE)	-	S	-	-	A	-	-	-
#4 (Paris, FR)	-	A	A	A	A	-	-	-
#5 (Oregon, US)	-	A	A	A	A	-	-	-
#6 (Tor)	-	A	A	A	A	-	-	-

A: Additional authentication factors requested
S: Security alert submitted

It‘s still not that easy...

Study two

Test all features with the IP threshold*

*Set IP one step below RBA threshold, set other features as “suspicious“ as possible

Results

Google

	Result
IP address	S
User agent	S
Language	-
Time	-
Resolution	S

	IP	UA	L	T	R
IP address		S	S	S	S
User agent	S		S	S	S
Language	S	S		-	S
Time	S	S	-		S
Resolution	S	S	S	S	

IP	UA	L	T	R	Result
X		X	X		S
	X	X	X		S
		X	X	X	S
X	X		X		A/C
X	X	X	X		A/C
X	X		X	X	A/C
X	X	X	X	X	A/C

A: Additional authentication factors requested
S: Security alert submitted
C: Critical security alert submitted

Results

LinkedIn

	Result
IP address	-
User agent	-
Language	-
Time	-
Resolution	-

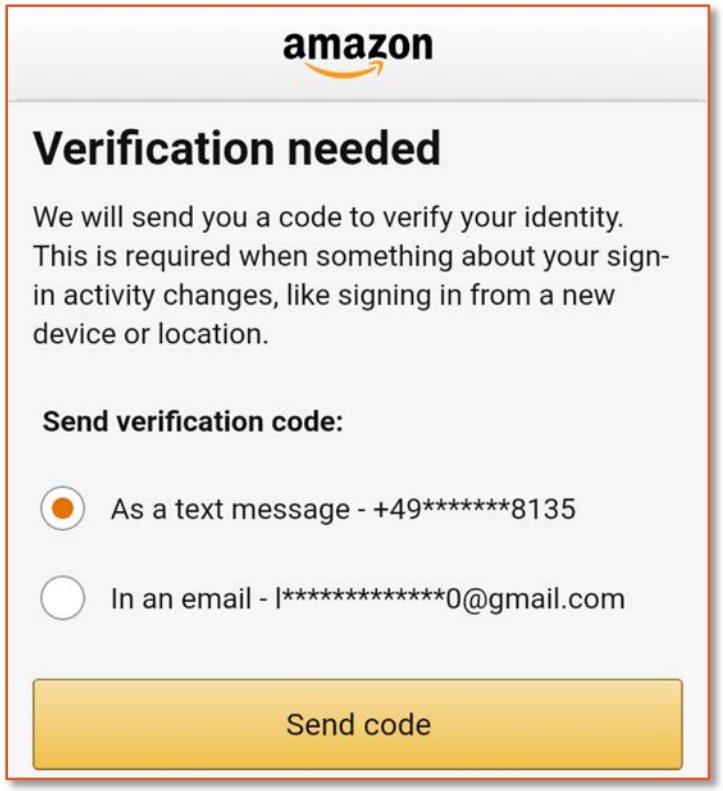
	IP	UA	L	T	R
IP address		A	A	A	A
User agent	A		-	-	-
Language	A	-		-	-
Time	A	-	-		-
Resolution	A	-	-	-	

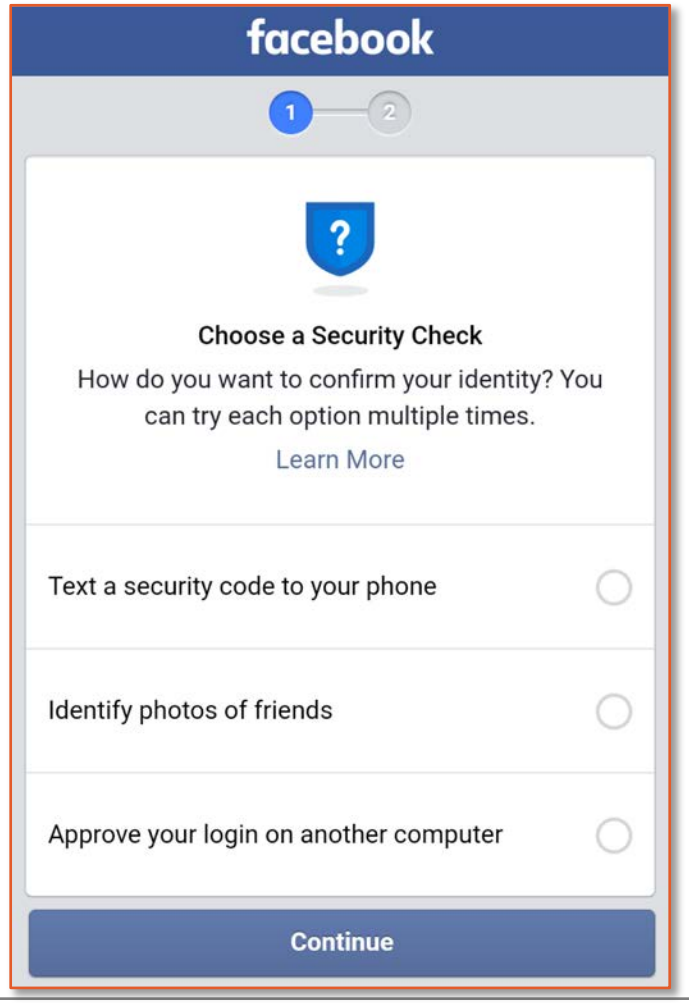
A: Additional authentication factors requested

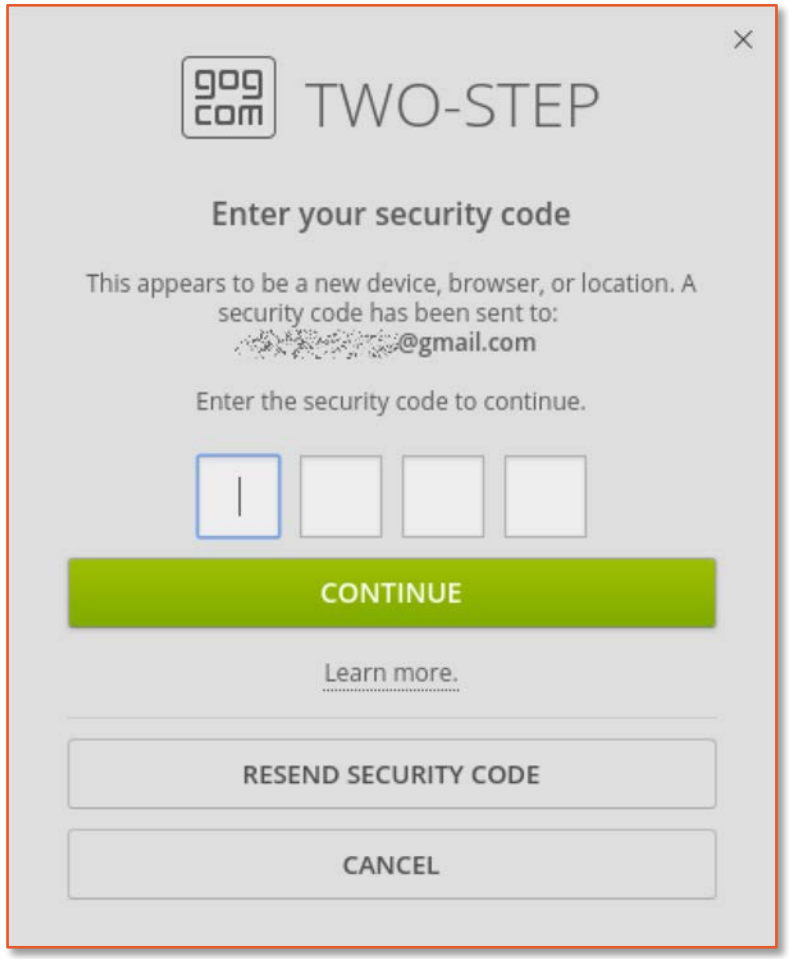
Results

Service	Used features and weightings
Amazon	IP address
GOG.com	IP address
Google	1. IP address 2. Time parameters 3. User agent string, display resolution
LinkedIn	1. IP address 2. User agent string, language, time parameters, display resolution

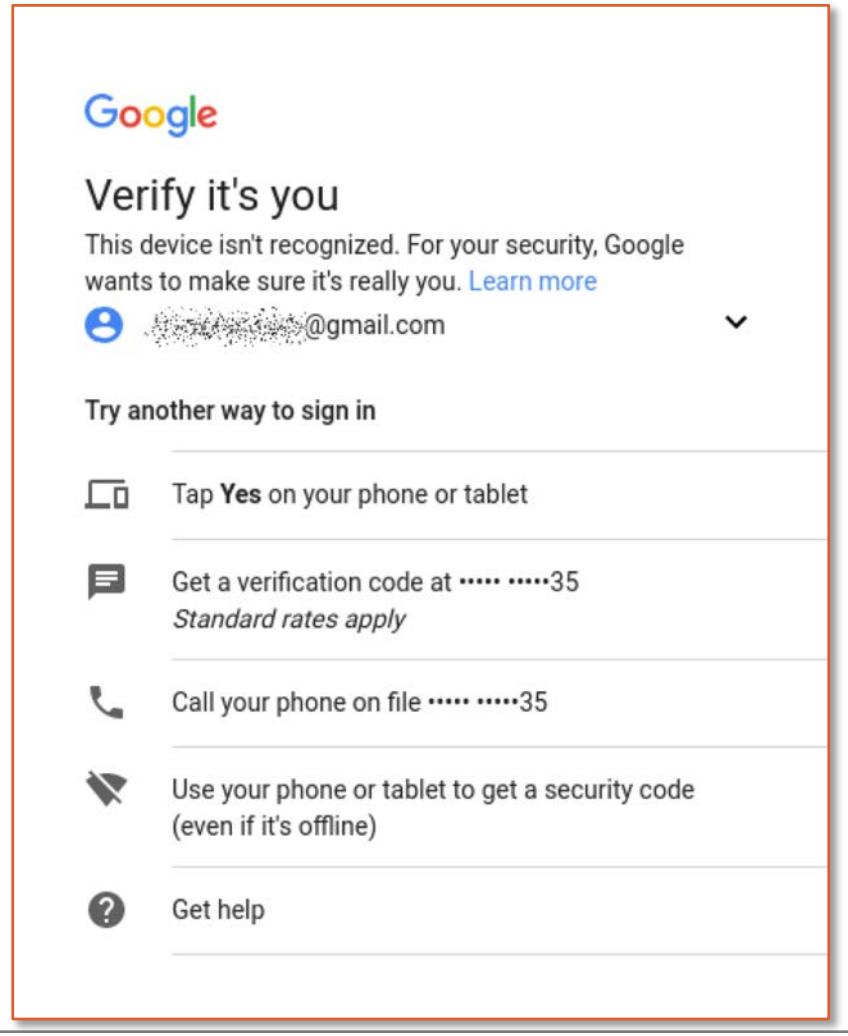
Results

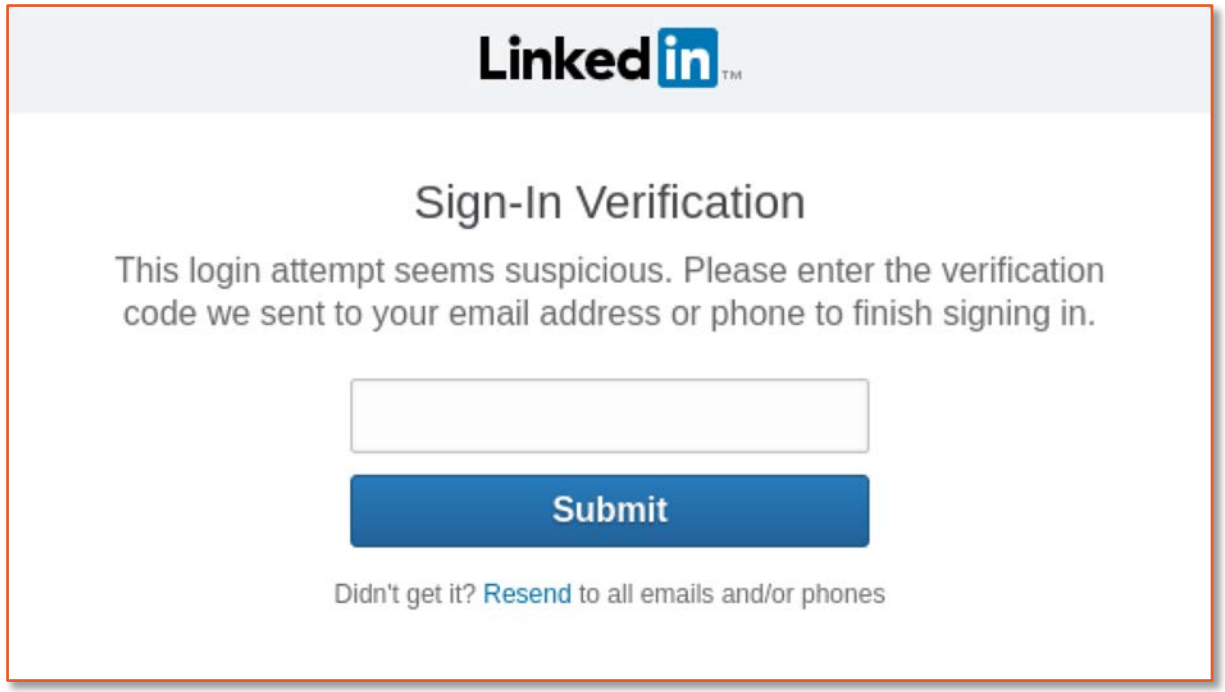

amazon
Verification needed
We will send you a code to verify your identity. This is required when something about your sign-in activity changes, like signing in from a new device or location.
Send verification code:
As a text message - +49*******8135
In an email - l**********0@gmail.com
Send code


facebook
1
2
Choose a Security Check
How do you want to confirm your identity? You can try each option multiple times.
Learn More
Text a security code to your phone
Identify photos of friends
Approve your login on another computer
Continue


gog com
TWO-STEP
Enter your security code
This appears to be a new device, browser, or location. A security code has been sent to:
@gmail.com
Enter the security code to continue.
CONTINUE
Learn more.
RESEND SECURITY CODE
CANCEL

Results


Google
Verify it's you
This device isn't recognized. For your security, Google wants to make sure it's really you. Learn more
@gmail.com
Try another way to sign in
Tap Yes on your phone or tablet
Get a verification code at ····· ·····35
Standard rates apply
Call your phone on file ····· ·····35
Use your phone or tablet to get a security code (even if it's offline)
Get help

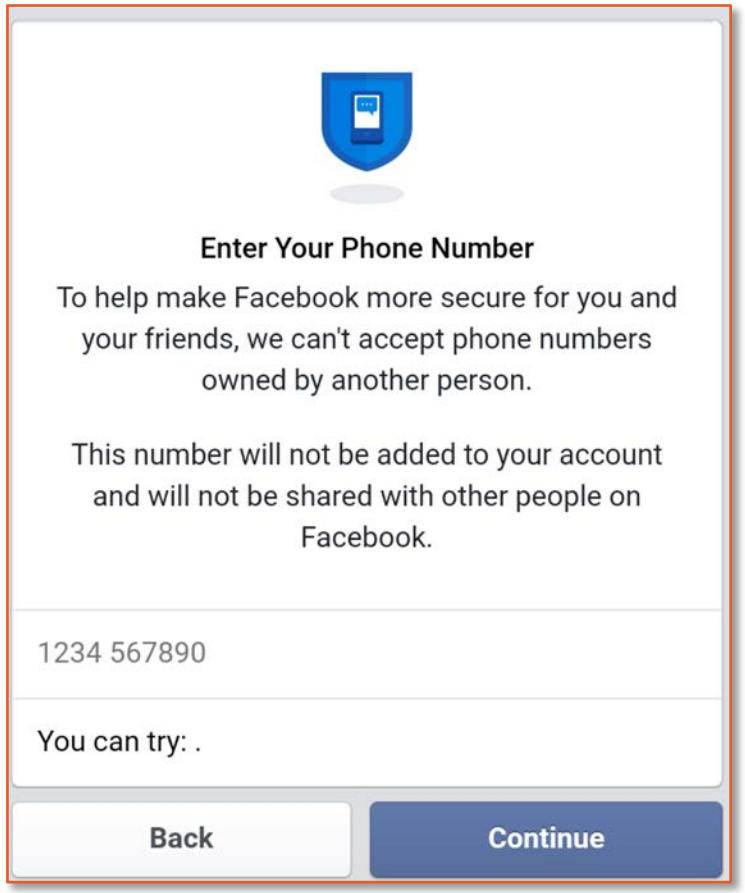

LinkedIn
Sign-In Verification
This login attempt seems suspicious. Please enter the verification code we sent to your email address or phone to finish signing in.
Submit
Didn't get it? Resend to all emails and/or phones

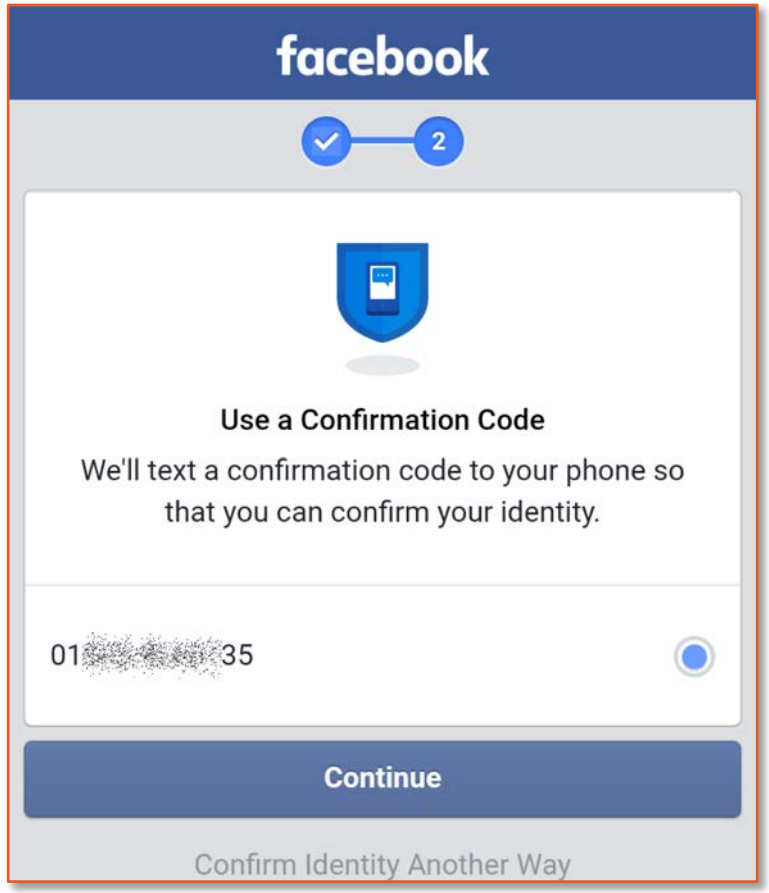
Results

Service	Requested authentication factors
Amazon	▪ Verification code (email*, text message)
Facebook	▪ Approve login on another computer ▪ Identify photos of friends ▪ Asking friends for help ▪ Verification code (text message)
GOG.com	▪ Verification code (email)*
Google	▪ Enter the city you usually sign in from ▪ Verification code (email, text message, app, phone call) ▪ Press confirmation button on second device
LinkedIn	▪ Verification code (email)*

*: Authentication factor was offered in all tested parameter variations

Privacy leak on Facebook



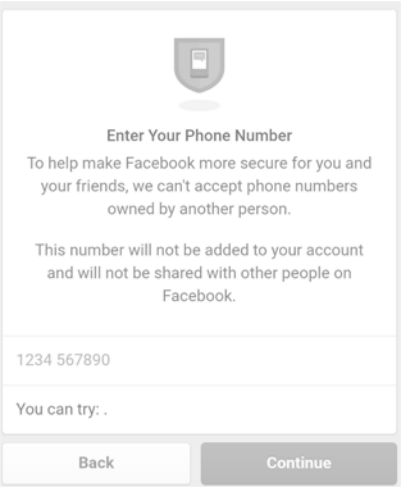


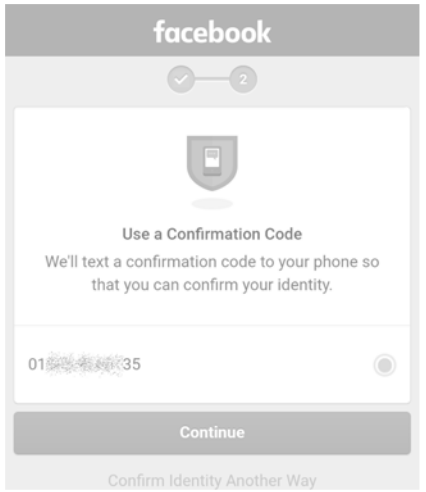
Privacy leak on Facebook

Responsible disclosure

Reported: September 4th, 2018

Fixed: September 6th, 2018





Conclusion

- First insights into RBA practices of big online services
- Intended to support developers, administrators and researchers
- Testing tool available as open source software*
- Interactive results and RBA models on website#

*https://github.com/das-th-koeln/HOSIT

#https://riskbasedauthentication.org

Thank you

riskbasedauthentication.org
das.th-koeln.de

stephan.wiefling@th-koeln.de

@swiefling